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**SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE**

(Autonomous Institution, Reaccredited by NAAC with 'A+' GRADE Approved by AICTE and Permanently affiliated to Anna University, Chennai)

**AUTONOMOUS EXAMINATIONS - NOV'2024**

**B.E – COMPUTER SCIENCE AND ENGINEERING**

**20CS2E48 - SOFTWARE TESTING**

**Answer ALL questions**

**Duration: 2 Hrs 30 min**

**Maximum: 70 Marks**

**PART - A**

**(5x 1 = 5 Marks)**

1. **CO1 In which phase of SDLC is the "Logical Design" phase performed?**
  - a) Requirement phase      b) Design phase
  - c) Development phase      d) Testing phase
  
2. **CO2 Identify Which of the following testing techniques involves testing without knowledge of the internal code or logic of the application?**
  - a) White-box testing      b) Black-box testing
  - c) Gray-box testing      d) Regression testing
  
3. **CO3 Define "Data-pool" in testing?**
  - a) A centralized repository for storing all test scripts
  - b) A collection of test data used for automated testing
  - c) A collection of user stories for functional testing
  - d) A summary of testing results from multiple test runs
  
4. **CO3 Identify Which of the following is typically included in an automated test procedure?**
  - a) Test case creation, execution, and result logging
  - b) Documentation of the test environment setup
  - c) Manual intervention for each test step
  - d) Identification of bottlenecks in the application

5. **CO4** Identify Which of the following is NOT a typical output of a software review?
- a) Defect logs and recommendations
  - b) Review meeting minutes and actions taken
  - c) A detailed user manual
  - d) A summary of software quality metrics

**PART - B**

**(5x 3 = 15 Marks)**

6. **CO1** List the main components of a Test Plan?
7. **CO1** Describe how does Testing in the Requirements Phase contribute to software quality?
8. **CO2** List the advantage of Automated Unit Testing over manual testing?
9. **CO3** Predict the role of Bottleneck Identification in performance testing?
10. **CO4** Differentiate "Peer Review" and a "Code Inspection."

**PART - C**

**(5 x 10 = 50 Marks)**

**Q.No: 11 Compulsory Question**

**Q.No: 12 to 17 Answer any FOUR Questions**

**Compulsory Question:**

11. **CO1** i) Analyze the role of testing in the Software Development Life Cycle (SDLC) and explain how it impacts the overall project success. (5)
- CO1** ii) Discuss the different types of software testing techniques (manual vs automated) and explain the advantages and disadvantages of each. (5)
12. **CO1** i) Discuss the challenges and best practices involved in Test Project Management, particularly in estimating test costs and duration. (5)
- CO1** ii) Discuss the importance of Test Requirements Specification. (5)
13. **CO2** i) You have been assigned the role of a Software Test Engineer to create a set of test cases for an e-commerce application. This application allows users to browse products, add items to their shopping cart, complete checkout, and make payments using various methods like credit/debit cards and digital wallets. The application also includes user login, registration, and account management features. The product catalog includes categories like electronics, clothing, and groceries. The (5)

application must be tested across various devices (desktop, mobile, and tablet) and browsers (Chrome, Firefox, and Safari). Your task is to show comprehensive test cases for the following features of the e-commerce application:

- User Registration and Login
- Product Search and Browse

**CO2 ii)** You are part of the QA team for a newly developed mobile banking app. (5)

The app includes the following features:

- User registration and login
- Viewing account balances and transaction history
- Transferring funds between accounts
- Bill payments
- Push notifications for transaction updates

Illustrate what types of tests would you include (e.g., functional testing, security testing, usability testing, etc.)? Why?

14. **CO2 i)** Explain White box testing, and how does it differ from Black Box testing? (5)

**CO3 ii)** Show how do test scripts differ from test plans, and why are both important in a testing strategy? (5)

15. **CO3 i)** Show how would you select which test cases to automate at the unit testing level (5)

**CO3 ii)** Differentiate "big bang integration" and "incremental integration." (5)

16. **CO4 i)** Compare and contrast formal and informal reviews in software testing. (5)

**CO4 ii)** Explain the role of walkthroughs in identifying issues in early stages of software development. (5)

17. **CO4 i)** List and describe the essential components of a software review plan. (5)

**CO4 ii)** Show how would you prioritize issues identified in a review report for a large-scale project? (5)

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